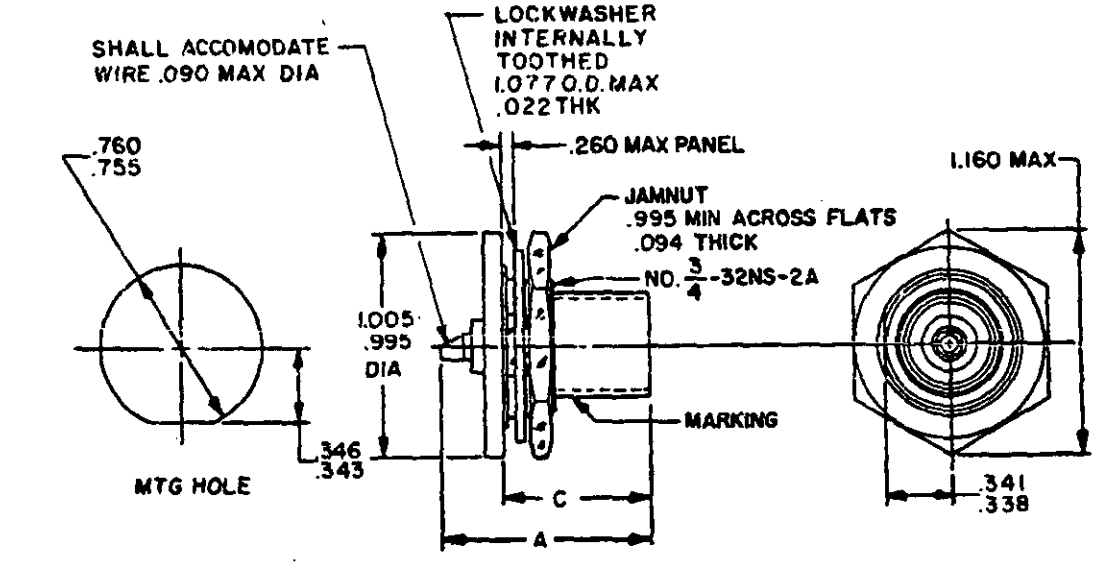
MILITARY SPECIFICATION SHEET

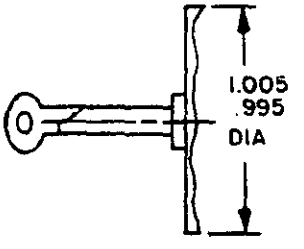
CONNECTORS, COAXIAL, RADIO FREQUENCY

SERIES SC (UNCABLED - RECEPTACLE, FEMALE, JAM NUT MOUNTED, REAR MOUNTED, CLASS 2)

The complete requirements for procuring the connector described herein shall consist of this document and the latest issue of Specification MIL-C-39012.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.





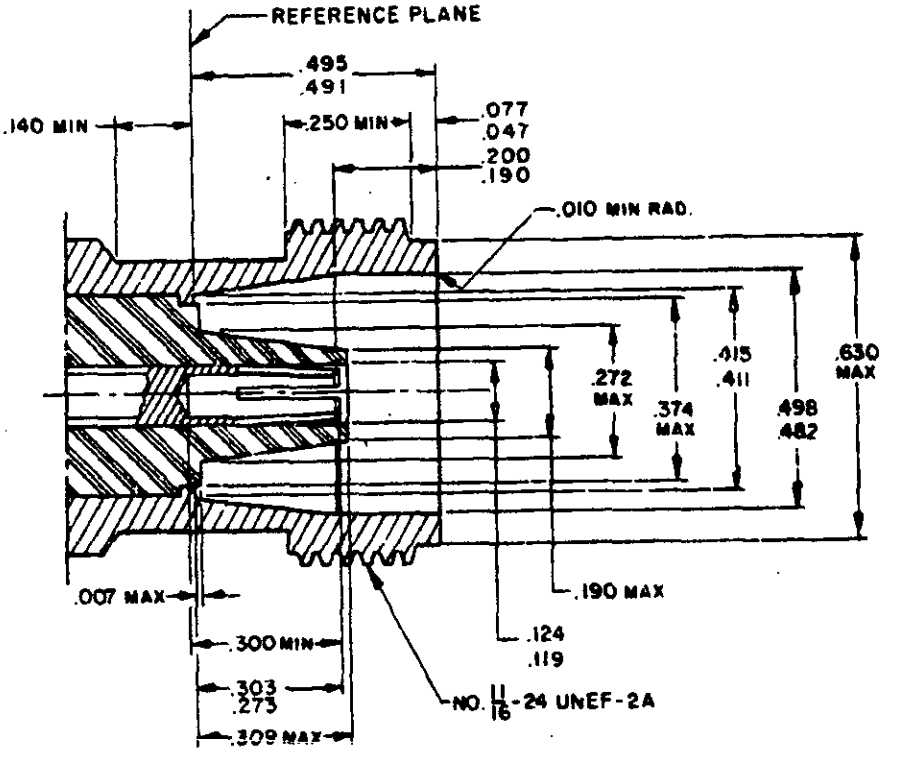
INCHES	MM	INCHES	MM
.022	.56	.755	19.18
.094	2.39	.760	19.30
.260	6.60	.995	25.27
.338	8.59	1.005	25.53
.341	8.66	1.077	27.36
.343	8.71	1.160	29.46
.346	8.79		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. For dimensions "A" and "C" see table I.
4. All undimensioned pictorial configurations are for reference purposes only.

FIGURE 1. General configuration.

FSC 5935

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INCHES	MM	INCHES	MM	INCHES	MM
.007	.18	.200	5.08	.411	10.44
.010	.25	.250	6.35	.415	10.54
.047	1.19	.272	6.91	.482	12.24
.077	1.96	.273	6.93	.491	12.47
.119	3.02	.300	7.62	.495	12.57
.124	3.15	.303	7.70	.498	12.65
.140	3.56	.309	7.85	.630	16.00
.190	4.83	.374	9.50		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. All undimensioned pictorial configurations are for reference purposes only.

FIGURE 2. Mating dimensions for female terminations.

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TABLE I. Part number and basic overall dimensions.

Part no.	Dim	Inches - millimeters 1/		Type of seal
		Minimum	Maximum	
M39012/43-0001	A	---	1.390(35.31)	Pressurized
	C	.915(23.24)	.975(24.77)	
M39012/43-0002	A	---	1.812(46.02)	Hermetic
	C	.980(24.89)	1.010(25.65)	

1/ Millimeters are in parentheses.

ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: 0 to 11,000 MHz.

Voltage rating: 1,000 volts rms maximum working voltage at sea level, 250 volts rms maximum at 70,000 feet.

Temperature rating: -65°C to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Force to engage and disengage:
Longitudinal force - Not applicable.
Torque - 6 inch pounds maximum.

Coupling proof torque: Not applicable.

Inspection conditions: Coupling torque - 6-10 inch pounds.

Mating characteristics:
See figure 2 for dimensions.
Center contact (female):
Oversize test pin - .098 diameter minimum, 16 microinch finish (non-closed entry contacts only).
Insertion depth - .094 minimum.
Number of insertions - 1.

Insertion force test - Steel test pin diameter .092 minimum.
Test pin finish - 16 microinches.
Insertion force - 2 lbs maximum.

Withdrawal force test: Steel test pin diameter .090 maximum.
Withdrawal force - 2 ounces minimum.
Test pin finish - 16 microinches.

Hermetic seal: Leakage shall not exceed 1×10^{-5} cc/sec of tracer gas at atmospheric pressure. Applicable to M39012/43-0002 only.

Leakage (pressurized connectors):
Connectors shall be mounted in mounting hole shown on figure 1 with mating end capped. Test applicable to mounting seal only. Air pressure - 30 psi. Duration - 30 seconds minimum, 2 minutes maximum.

Insulation resistance: Method 302, test condition B, MIL-STD-202. 5,000 megohms minimum.

Center contact retention: 6 lbs minimum axial force.

Corrosion (salt spray): Method 101, test condition B, MIL-STD-202.

Voltage standing wave ratio (VSWR):
Not applicable.

Swept frequency VSWR test setup:
Not applicable.

Connector durability:
500 cycles at 12 cycles/minute maximum. The connector shall meet mating characteristics and force to engage and disengage requirements.

Contact resistance: In milliohms maximum.

	Initial	After environment
Center contact	1.0	1.5
Outer contact	.15	Not applicable
Braid to body	Not applicable	Not applicable

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Dielectric withstanding voltage: Method 301 of MIL-STD-202. 3,000 volts rms at sea level.

Vibration, high frequency: Method 204, test condition B, MIL-STD-202.

Shock: Method 213 of MIL-STD-202, test condition I.

Temperature cycling: Method 102, test condition C, MIL-STD-202.

Thermal shock: Applicable to -0002 only.

Moisture resistance: Method 106 of MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level:
Altitude - 70,000 feet.
Voltage - 750 volts rms minimum.

RF high potential withstanding voltage:
Voltage and frequency - 2,500 volts rms at 5 MHz.
Leakage current - Not applicable.

Cable retention force: Not applicable.

Coupling mechanism retention force: Not applicable.

RF leakage: Not applicable.

Insertion loss: Not applicable.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MI, MU, EL
Navy - OS
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - AT, AV, ME
Navy - AS, MC, SH
Air Force - 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5935-1696-9)

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Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-C-39012/43A, CONNECTORS, COAXIAL, RADIO FREQUENCY SERIES SC (UNCAPLED - RECEPTACLE, FEMALE, JAM NUT MOUNTED, REAR MOUNTED, CLASS 2)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

FSC-FM 1068-68

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